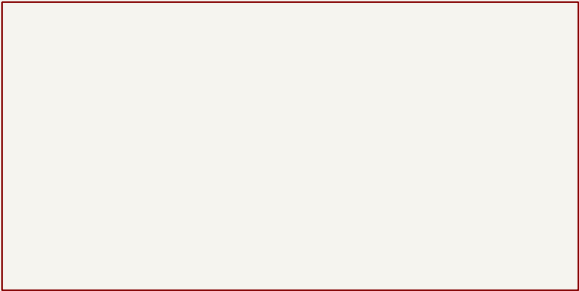


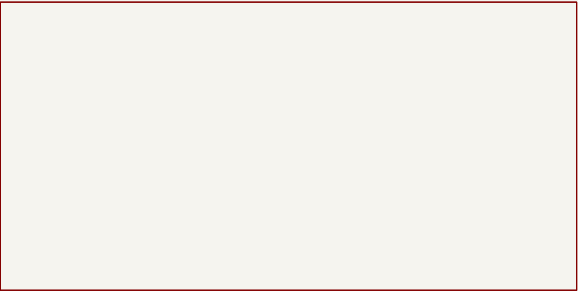
	1	2	3	4	5	6
A						A
B						B
C						C
D						D
	1	2	3	4	5	6

USB and protected power



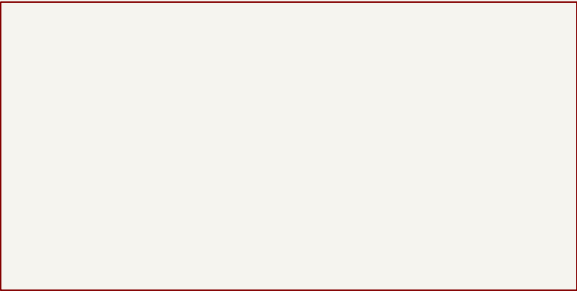
File: sheet1.kicad_sch

555 hardware timer



File: sheet2.kicad_sch

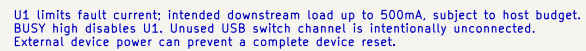
Supply and sequencing



File: sheet3.kicad_sch

Sheet: /		
File: usb-replug.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6	Id: 1/4	

Ground and cable shields stay connected. Both data lines and device VBUS are switched.



Sheet: /USB and protected power/
File: sheet1.kicad_sch

Size: A4	Date: 2026-09-11
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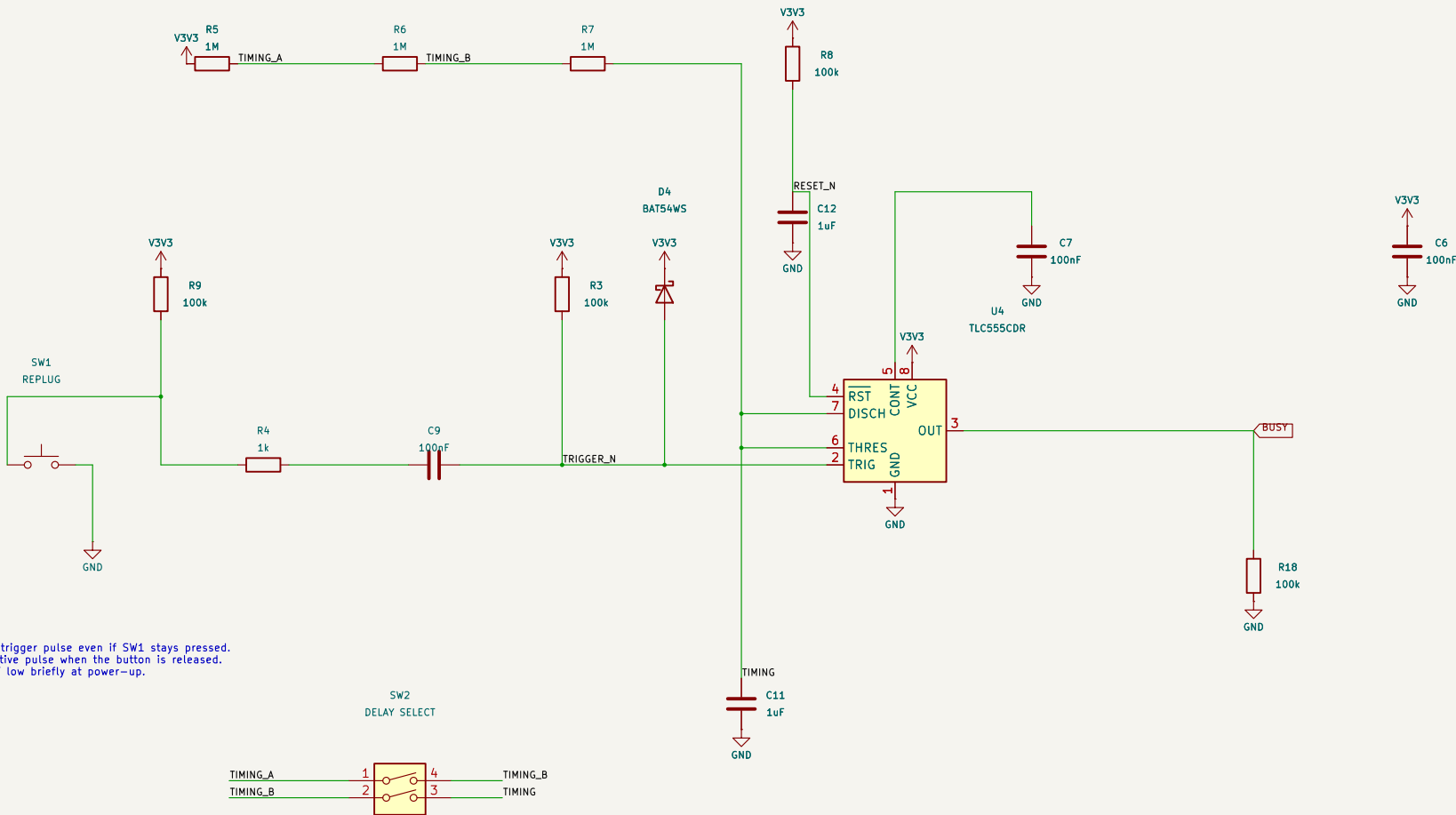
SIZE: A1	
KiCad E.D.A. 10.0.6	

Rev: B DRAFT

Id: 2/4

APPROXIMATE DELAY / NO FIRMWARE

Press to disconnect. Release is not required for automatic reconnection.



C9 creates a short trigger pulse even if SW1 stays pressed.
D4 clamps the positive pulse when the button is released.
R8/C12 hold RESET low briefly at power-up.

SW2: both OFF ~3s; either ON ~2s; both ON ~1s.
Change switches while idle. RC nominal: 1.10 / 2.20 / 3.30s.
Capacitance tolerance and DC bias reduce timing accuracy.

USB REPLUG		
Sheet: /555 hardware timer/ File: sheet2.kicad_sch		
Title: CMOS 555 timer – approximately 1 / 2 / 3 seconds		
Size: A4	Date: 2026-09-11	Rev: B DRAFT
KiCad E.D.A. 10.0.6		Id: 3/4

POWER RETURNS FIRST, THEN USB DATA

U2
TLV75533PDBVR

VBUS_HOST

C1
1uF

GND

V3V3

C2
1uF

GND

U6
SN74LVC2G14DBVR

1A_(Schmitt) 1Y_inverted

2A_(Schmitt) 2Y_inverted

GND

VCC

BUSY

GND

V3V3

CONNECT_REQ

DATA_OFF

D3
BAT54WS

FAST_OFF

R16
1k

R15
47k

RECONNECT_RC

C10
1uF

GND

Q1
BSS138

DISCHARGE

R17
1k

VBUS_DEVICE

BUSY

USB REPLUG

Sheet: /Supply and sequencing/
File: sheet3.kicad_sch

Title: Always-on supply, reconnect sequencing and discharge

Size: A4 Date: 2026-09-11 Rev: B DRAFT

KiCad E.D.A. 10.0.6 Id: 4/4

POWER RETURNS FIRST, THEN USB DATA

U2
TLV75533PDBVR

VBUS_HOST

C1
1uF

GND

V3V3

C2
1uF

GND

U6
SN74LVC2G14DBVR

1A_(Schmitt) 1Y_inverted
2A_(Schmitt) 2Y_inverted
GND
VCC

BUSY

GND

V3V3

CONNECT_REQ

D3
BAT54WS

FAST_OFF

R16
1k

R15
47k

DATA_OFF

RECONNECT_RC

C10
1uF

GND

Q1
BSS138

DISCHARGE

VBUS_DEVICE

R17
1k

BUSY

USB REPLUG

Sheet: /Supply and sequencing/
File: sheet3.kicad_sch

Title: Always-on supply, reconnect sequencing and discharge

Size: A4 Date: 2026-09-11 Rev: B DRAFT

KiCad E.D.A. 10.0.6 Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
Title: Always—on supply, reconnect sequencing and discharge		
Size: A4	Date: 2026-09-11	Rev: B DRAFT
KiCad E.D.A. 10.0.6		Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
Title: Always—on supply, reconnect sequencing and discharge		
Size: A4	Date: 2026-09-11	Rev: B DRAFT
KiCad E.D.A. 10.0.6		Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
Title: Always—on supply, reconnect sequencing and discharge		
Size: A4	Date: 2026-09-11	Rev: B DRAFT
KiCad E.D.A. 10.0.6		Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
Title: Always—on supply, reconnect sequencing and discharge		
Size: A4	Date: 2026-09-11	Rev: B DRAFT
KiCad E.D.A. 10.0.6		Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
Title: Always—on supply, reconnect sequencing and discharge		
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KiCad E.D.A. 10.0.6		Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
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KiCad E.D.A. 10.0.6		Id: 4/4

Sheet: /Supply and sequencing/ File: sheet3.kicad_sch		
Title: Always—on supply, reconnect sequencing and discharge		
Size: A4	Date: 2026-09-11	Rev: B DRAFT
KiCad E.D.A. 10.0.6		Id: 4/4